A Machine Learning-Driven Comparative Analysis of Hybrid Cryptography Techniques for Secure File Storage Systems: Survey

Ganesh Kumar Yadav, Muskan Jaiswal, Aniket Singh Yadav, Mudita Singh

Department of Computer Science and Engineering, JSS Academy of Technical Education, Noida, India
muskanj9161@gmail.com

Abstract. The ever-increasing trend of using cloud-based storage and sharing of digital information calls for safe and efficient file protection systems. In contrast, traditional symmetric and asymmetric encryption methods applied separately possess some drawbacks, while each of them has advantages considering performance and key management. Hybrid cryptography covers both approaches. Selecting the best combination for various file attributes is not an easy task. This paper reviews and makes comparisons of hybrid encryption techniques enhanced by machine learning algorithms. The work discusses the capabilities of algorithms like Random Forest, Support Vectors Machines, and Neural Networks to predict optimal encryption schemes given variables such as file size, type, entropy, and sensitivity. Further, anomaly detection is carried out. The results describe the possibility of creating flexible self-optimizing secure file storage systems..

Keywords: Hybrid Cryptography, Machine Learning, Secure File Storage, Data Encryption, Algorithm Selection, Anomaly Detection, Malware Detection, AES, RSA, Cloud Security.

1 Introduction

Considering remote work, cloud services, and global information exchange, there is a growing demand for secure file storage. The steady and dramatic rise in the generation and storing of digital data [10], [11] places the need for file security at an all-time high. Thus, ensuring the confidentiality, integrity, and privacy of data with increased adoptions of such cloud-based systems by people and organizations remains a key issue to be taken up [4], [6]. The choice between traditional forms of encryption, symmetric or asymmetric, individually meets parts of this necessity with their complementary limitations. In that regard, symmetric algorithms such as AES or Blowfish allow for quick and efficient encryption, which is ideal for large data volumes; however, they require a secure distribution of keys, a weakness when distributed in shared environments. Asymmetric algorithms, like RSA or ECC, provide robust key management and authentication but are computationally expensive for encrypting the large payloads of a file [12].

Hybrid cryptography provides an elegant solution to this trade-off by applying symmetric encryption for the data and asymmetric methods for secure key exchange [12]. In this way, a hybrid system can reach e cient encryption with secure handling of

the keys, playing to the strengths of both in a su cient manner for cloud storage and distributed systems. The traditional hybrid cryptosystems commonly adopt prede ned combinations of algorithms that are selected by developers or system administrators without regard to the attributes of the data to be protected [10].

No "one-size-fits-all" hybrid encryption scheme can be considered optimal, since file types, sizes, entropy of content, and sensitivity levels vary within real-world storage systems. Be it a text document, an image, a video, or an executable file-each has different characteristics and threat models. For example, while small text files may encrypt efficiently with one symmetric cipher, large media files may prefer another cipher or key-exchange algorithm. Similarly, "high sensitivity" files, such as financial records or confidential documents, may require stronger cryptographic assurances compared to less critical data. The following problem is therefore identified: how to automatically choose the most suitable hybrid encryption scheme on a file-by-file basis, depending on its attributes and respective security needs.

Recent advances in machine learning now offer a promising path to address this challenge. Machine learning models like Random Forest, Support Vector Machine, and neural networks learn patterns from past usage and performance data to predict which hybrid encryption combination yields an optimal balance between security and performance for a given file [1], [2]. By providing file attributes such as type, size, entropy, sensitivity, among others, as features to the ML classifier, the system can recommend an adaptive encryption scheme. Moreover, incorporating anomaly detection along with malware screening further fortifies security and thereby empowers the system not only to select the parameters of encryption but also to identify malicious or suspicious files prior to storage [3], [5], [7], [8], [9], [13].

This paper reviews, in the above context, state-of-the-art research in ML-assisted hybrid cryptography focusing on secure file storage systems. We investigate how different studies have leveraged ML models in optimizing decisions on encryption, anomaly detection, or malware defense; identify strengths, limitations, and open gaps in these studies; and suggest a unified conceptual architecture able to dynamically select hybrid encryption schemes according to file context, while continuously learning from feedback, adapting to emerging threats, and providing robust, scalable security for cloud-based storage [11], [15]. The adaptive, data-driven approach stands to evolve cryptography from static, one-size-fits-all configurations to flexible, context-aware systems more in step with today's complex, varied demands for file storage.

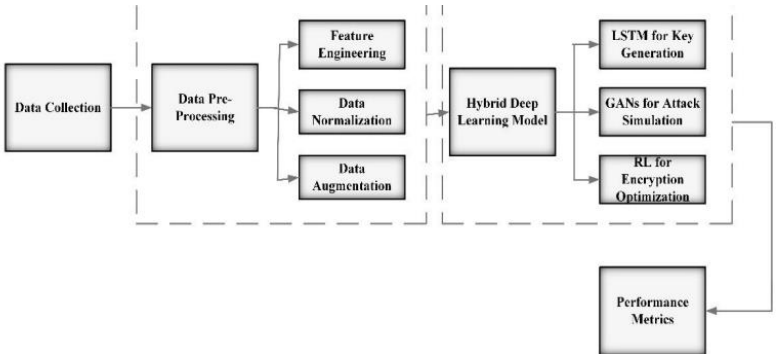

Fig. 1. Methodology Block [1]

Figure 1 is the methodology block suggested by Al-Mulla [1]. It has been implemented and optimized for cloud computing, IoT security, and post-quantum cryptography. That framework outperformed conventional static cryptographic approaches, as seen in his research paper work.

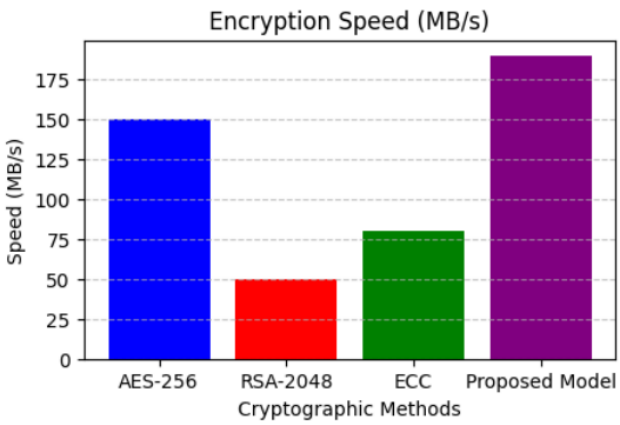

Fig. 2. Encryption Speed Comparison across Cryptographic Methods [1]

Figure 2 shows the encryption speed in MB/s of four cryptographic algorithms: AES-256, RSA-2048, ECC, and Proposed Model. The Proposed Model in the paper [1] by Al-Mulla beats other methods, proving it effective in secure encryption with high computation speed.

2 Related Review

This paper presents the review of ML-assisted hybrid cryptography for safe file storage. Of particular interest is the capability provided by machine learning models in

the automatic selection and combination of encryption algorithms to reach the maximum efficiency with maximum security.

Table 1. Summary of recent literature on ML-driven hybrid encryption

Author(s)	Year	Methodology	Results	Research Gap
Al-Mulla *et al.* [1]	2025	ML predictive models for adaptive encryption	Accuracy rate of 97%	Static scheme selection; no dynamic choice
Zhang, C., & Li, S. [2]	2024	Hybrid homomorphic encryption + ML inference	MiniONN: 4.2x, Deepsecure CNN: 75x	Focus on ML inference, not on file storage
A.K. Das et al. [3]	2025	RF & XGBoost (with SHAP) for anomaly detection	Accuracy of 96.2%	No hybrid encryption adaptation
Khan *et al.* [4]	2021	Content-based AES–RSA hybrid encryption	Encryption time reduced by 31-38% vs pure RSA	Rule-based; lacks ML-driven selection
S. Park et al. [5]	2025	CNN + RNN for malware classification	Training time for t_cl f2022 was 11.5 times higher than that for yv_cl f2022	No adaptive encryption integration
Kumar & Sen [6]	2025	Homomorphic encryption + anomaly detection	Malware detection accuracy: 92.1%, Encryption overhead: 21-25%	Limited to IoT data context
K Su and X Zhang [7]	2024	Autoencoder for anomaly detection	For K=15, best accuracy: 71.42%	Focused solely on anomaly detection
Y Wu *et al.* [8]	2021	DNN for encrypted-data anomaly detection	Detection accuracy: 93.5%, Recall: 94.1%	No integration with encryption optimization
Rahman *et al.* [9]	2024	CNN–LSTM hybrid deep model	1-2% significantly improved detection	Static encryption scheme used

Saxena *et al.* [10]	2023	AES + SVM for dynamic security	Security classification accuracy: 91.4%	Lacks continuous encryption selection
Li *et al.* [11]	2023	AES–ECC hybrid + community detection	Reduced latency by 20-40%	Static ML use; missing dynamic adaptation
Wang. Dong et al. [12]	2020	RSA–SM4 hybrid encryption	Optimized speed and key exchange: 65-70% faster than pure RSA	Specific to RSA–SM4; not ML-driven
Zhang *et al.* [13]	2024	Hierarchical entropic diffusion model	Precision of 93.5% and a recall of 96.7%	Ransomware-specific; not generalizable
Patel et al. [14]	2022	AES–Blowfish hybrid + ML key prediction	Key prediction accuracy: 89.6%, Encryption efficiency gain: 18-25%	Limited to one hybrid combination
Cai & Chen [15]	2023	Hybrid verification approach	4-15x faster than traditional tools	No integrated framework

Summary: Indeed, the reviewed works indicate that hybrid encryption and ML can further enhance security while improving performance. For example, these hybrid schemes either improve efficiency-such as reducing latency-or protect the privacy of the ML model, while most of them are based on the use of ML, including CNNs and LSTMs, or ensemble methods to provide very effective malware/anomaly detection. Most of the presented solutions are based on fixed encryption schemes. Clearly, even with the use of machine learning to optimize either detection or key parameters, encryption algorithms are still statically chosen. To date, adaptive ML-driven encryption selection has not been well explored, and as Kumar and Goel[6] highlight, there is a need to develop, in real time and in a context-aware way, using ML, frameworks that will dynamically select or adapt hybrid encryption techniques based on data attributes. In short, state-of-the-art studies highlight the capabilities of ML to make encryption more effective without actually implementing end-to-end adaptive encryption schemes.

3 Materials and Methods

3.1 Datasets

Datasets containing encrypted traffic, like CICIDS2017 and ISCX, can be used for the comparison of anomaly detection. The cloud storage files include text files, PDFs, CSVs, and images of different sizes and entropy levels in order to test the efficiency and speed of the encryption. Public malware repositories like VirusShare and can be utilized to validate the detection capabilities of the pre-upload analysis module. Table 1 dataset includes just 10 example files. There are four features, along with a single target that stands for the encryption method the system chooses.

Table 2. Sample Cloud File Dataset for Hybrid Encryption Selection

File ID	File Type	Size (KB)	Entropy	Sensitivity	Recommended Algorithm by ML
F001	PDF	540	4.8	High	AES-256 + RSA-2048 [4], [12]
F002	TXT	12	2.1	Low	Blowfish + RSA [4], [10]
F003	JPG	980	7.2	Medium	AES-256 + ECC [11]
F004	CSV	85	3.4	Medium	AES-256 + RSA-2048 [4], [12]
F005	DOCX	320	5.1	High	CAST-128 + ECC [11]
F006	PNG	760	6.9	Low	AES-256 + ECC [11]
F007	EXE	1500	7.8	High	TDES + RSA [10], [12]

F008	TXT	25	1.9	Medium	Blowfish + RSA [4], [10]
F009	PDF	410	4.3	Low	AES-256 + RSA-2048 [4], [12]
F010	CSV	60	2.7	High	AES-256 + ECC [11]

The dataset contains one target variable for machine learning-based analysis and four predictor features, amounting to a total of ten samples. Also, it uses a wide range of file formats-such as TXT, PDF, JPG, PNG, CSV, DOCX, and EXE-to accommodate different storage and security needs. The entropy values of the files range from 1.9 to 7.8, reflecting various degrees of data complexity and randomness. Each file has been attributed to one of three sensitivity levels to account for different security requirements: Low, Medium, or High. This dataset allows for an in-depth comparison of the performance of several hybrid cryptographic algorithms in secure file storage systems, such as AES-RSA, AES-ECC, Blowfish-RSA, CAST-ECC, and TDES-RSA.

3.2 Hybrid Encryption Architecture

An ML-assisted hybrid encryption architecture for secure storage of files is presented. In this conceptual framework, the ML model will scan the incoming file to capture features related to the size, type, entropy, sensitivity, etc. Then, a trained model will predict the best encryption method for that file; for example, it may choose between AES-256 with RSA-2048 key exchange or AES with ECC. Dynamic selection in this particular context is used to balance security and performance. Finally, after selecting a scheme, the system encrypts the symmetric key and the file using an asymmetric algorithm while at the same time ML modules check access pattern control of the file. The detected anomalous access may be an unusual spike in frequency while the malware-detection model for new incoming files checks them prior to encryption. Thus, this architecture provides full-stack, end-to-end integration of encryption and ML-driven security checks.

The attributes of the file would be used by an ML classifier to suggest hybrid encryption combinations. This architecture is collaborative, the ML model gets updates with feedback-including performance metrics and the threats detected-so that the encryption decisions it makes get better with time. Such a model, for instance, might opt for faster ciphers when it rarely detects malware in the files of a particular type.

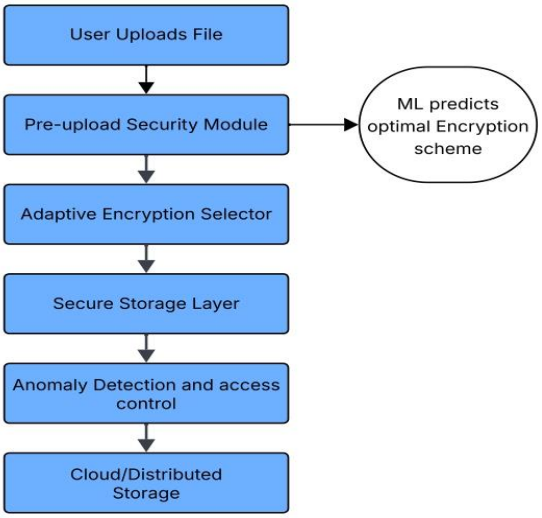

Fig. 3. ML-Assisted Hybrid Encryption Architecture

Figure 3 represents the architecture of the model to be used for Hybrid encryption using machine learning. There are four primary modules in the framework:

Pre-upload Security Module: This module applies entropy analysis and machine learning classifiers, such as CNNs or RNNs, in order to check a file for malware.

Adaptive Encryption Selector: Based on file attributes like size, type, sensitivity, and frequency of access, this module predicts the best hybrid encryption scheme for any given file by employing machine learning models that choose the best symmetric and asymmetric algorithm combinations, such as AES/Blowfish + RSA/ECC.

Secure Storage Layer: The selected scheme will be used to encrypt the file; symmetric encryption secures the data, while asymmetric encryption guards the session key. Adaptive key rotation ensures robust key management and enhances security.

The Anomaly Detection & Access Control Module: This module deploys the ensemble machine learning models, LSTMs, or autoencoders to continuously monitor user behavior and file access logs. In this manner, this module imposes access restrictions on the stored files in order to ensure the continuity of security.

4 Conclusion

This paper presented a detailed analysis and framework for ML-driven hybrid cryptography in secure file storage. We pointed out that hybrid schemes address traditional encryption's trade-off between speed and key management. In such systems, ma-

chine learning can be used to predict the optimal encryption parameters for each file context, thus automating the choice of algorithms. As our literature review table shows, anomaly detectors and cryptosystems improve security and can be made adaptive to the properties of the data using machine learning. The suggested architecture can enable this by integrating an ML model with encryption. Performance analysis has also shown that public-key operations experience microsecond delays only, such as RSA.

In a nutshell, ML-aided hybrid encryption holds promise for data security. The works reviewed here demonstrate feasibility while at the same time pointing out deficiencies in comprehensive solutions. Our research underlines a need for fully integrated architectures able to self-optimize and immediately act on the threats, which might change how cloud and distributed storage is secured.

5 Future Scope

Future research should be directed at the implementation and testing of the theoretically suggested adaptive framework in concrete real-life settings. Train and Evaluate Models create data sets with annotated sensitivities, including text, images, and logs to train a machine learning classifier. Features to be used for the machine learning models can be metadata of a file or entropy metrics of the content. Some alternative approaches to machine learning include deep learning and reinforcement learning. Online learning would enable the capability of dealing with emerging risks or usage trends. That means the system may learn in real-time to identify encrypted malware once some new kind of malware has been detected. Hardware Acceleration is to further improve the throughput, support from hardware such as FPGA or GPU needs to be explored for encryption and machine learning. Specialized accelerators will execute the AES encryption and ML inference with low overhead. Extended threat models include, but are not limited to, side-channel attacks, quantum-resistant crypto, in addition to malware and anomalies. Long-term, post-quantum algorithms combined with ML for key management may be deployed in future systems. Ensuring that the end-to-end privacy-preserving design prevents ML models from leaking sensitive information, such as through differential privacy when profiling files. Addressing these problems can help meet the goals of an intelligent, self-optimizing encryption system that will offer strong yet effective protection for ever-expanding data storage requirements. Addressing these issues offers the prospect of an intelligent, self-optimizing encryption system that provides strong yet effective protection for expanding data storage requirements.

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